Kautik Rehani

BS–MS student, Physics major and Mathematics minor, IISER Pune
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Education

Indian Institute of Science Education and Research, Pune Expected July 2028
BS–MS Dual Degree; Physics major, Mathematics minor

Research Experience

Long-term Project Student, IISER Pune Aug 2025–present
Supervisor: Prof. Sudarshan Ananth

- Worked through three-point interaction terms for general spin and the quartic gravity interaction in terms of the two graviton helicity states in light-cone gauge.
- Worked out the full Yang-Mills Lagrangian in light-cone gauge; investigated higher-order terms in closed-form gravity expressions with Dr. Nipun Bhave.
- Current reading: Color-kinematics duality and $\mathcal{N} = 8$ supergravity in physical modes.

Summer Intern, NCRA-TIFR, Pune May - Jun 2025
Supervisor: Prof. Ishwara Chandra

- Worked on code to identify changing-look AGN candidates in large datasets, using SDSS Data Releases 16 and 17.
- Studied changing-look AGNs and their connection to black-hole accretion.

Summer Intern, TIFR, Mumbai May - Jul 2024
Supervisor: Prof. Shadab Alam

- Worked on QSO spectral stacking within the broader project *QSOs as tracers of our Universe.*
- Studied spectral analysis and explored black-hole kick-velocity distributions.

Relevant Coursework

QFT-1, Prof. Diptimoy Ghosh
Classical field theory, symmetries, free-field quantisation, interacting theories and scattering.

Differential Geometry and Lie Groups, Prof. Tejas Kalelkar
Smooth manifolds, differential forms, integration, Lie groups, Lie algebras and group actions.

Gravitation and Quantum Entanglement, Prof. Sunil Mukhi
Entanglement in QFT, modular Hamiltonians, black-hole thermodynamics and holographic entropy.

Technical Skills & Languages

Programming and computing: Python, TensorFlow; basic Mathematica, SageMath and Gnuplot.
Tools: LaTeX, GitHub/GitLab, Linux/Ubuntu, Jupyter Notebook, Anaconda.
Languages: English and Hindi (fluent); French (basic reading and writing).

Achievements

NEST 2023: All India Rank 30.
KVPY-SA 2021: All India Rank 1557.

Activities & Involvement

Akashganga: Astronomy Club member; telescope-building and cosmology workshops.
Disha: Volunteer in science outreach for local school students.
Smashers: Founded an institute sports community with approximately 200 members.